HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Provided technical input and leadership for **strategic power system analysis** projects, including short circuit and power flow studies for enterprise clients.
- Translated client requirements into actionable deliverables for **power systems engineering** projects, ensuring adherence to project specifications and client expectations.
- Prepared detailed technical and non-technical reports for complex power system analyses, including **arc flash and electrical risk assessments**.
- Conducted in-depth **power systems analysis**, focusing on protection coordination studies and cable sizing calculations per IEC standards.
- Managed client expectations through effective communication and timely responsiveness during remote client meetings and project updates.
- Developed and mentored junior engineers on **power system study software** such as ETAP, enhancing team capabilities and project execution.
- Containerized a Python-based power systems analysis tool using **Docker** and deployed it to an **AWS EC2** instance, enabling scalable on-demand computations.

SKILLS

Power Systems Engineering – Power System Analysis, Short Circuit Analysis, Power Flow, Protection Coordination, Arc Flash, Electrical Risk Assessment, Grid Connection, Transient Stability, Harmonics, Power Quality, Earthing, Cable Sizing, IEC 60502-2, IEC 60364-5-52

Power System Study Software – ETAP, SKM, EasyPower

Programming & Scripting – Python, Java, SQL

Cloud & DevOps – AWS EC2, Docker, Jenkins, Git

Electrical Protection – Unit Protection Settings, Bus Differential (87B), Transformer Differential (87T), Restricted Earth Fault, Directional Protection (67)

PROJECTS

Power System Reliability Assessment Tool (HCL Technologies)

- Developed a Python application to perform electrical systems reliability assessments, integrating with client-provided data files.
- Implemented algorithms for calculating redundancy and failure rates based on component specifications and historical data, improving analysis efficiency by **30%.**
- Designed the application's data processing pipeline using **pandas** and generated summary reports for stakeholder review.

Embedded Generation Connection Analysis Framework (HCL Technologies)

- Built a Python framework to automate analysis for embedded generation connections, focusing on static transient stability and transformer inrush calculations.
- Integrated calculations for harmonics and flicker, providing early identification of potential grid impact issues for utility clients.
- Engineered the framework to accept standardized input formats, reducing manual data entry time by approximately **40%** for routine connection studies.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)